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Inventor(s)	Chen; Chin-Liang et al.

Semiconductor package and methods of fabricating a semiconductor package

Abstract

A semiconductor package provided herein includes a package substrate and a semiconductor device. The package substrate includes a redistribution structure, an interconnect structure bonded to the interconnect structure and an insulation material laterally surrounding the interconnect structure, wherein the redistribution structure has a reduced structure and the insulation material fills the reduced structure. The semiconductor device is bonded to the package substrate. In addition, a method of fabricating a semiconductor package is also provided and includes a precut process forming the reduced structure in the redistribution structure of the package substrate.

Inventors: Chen; Chin-Liang (Kaohsiung, TW), Hou; Hao-Cheng (Hsinchu, TW), Liang; Yu-Min (Taoyuan, TW), Cheng; Jung-Wei (Hsinchu, TW), Wang; Tsung-Ding (Tainan, TW)

Applicant: Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)

Family ID: 89577087

Assignee: Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)

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Primary Examiner: Sofocleous; Alexander

Assistant Examiner: Sylvia; Christina A

Attorney, Agent or Firm: JCIPRNET

Background/Summary

BACKGROUND

(1) Semiconductor packages are used in a variety of electronic applications, such as personal computers, cell phones, digital cameras, and other electronic equipment. In terms of the packaging used for integrated circuit components or semiconductor dies, one or more packaged semiconductor devices such as chip packages are generally bonded to a circuit carrier (e.g., a system board, a printed circuit board, or the like) for electrical connections to other external devices or electronic components. Although the existing semiconductor package has been generally adequate for their intended purposes, it has not been entirely satisfactory in all respects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1A and FIG. 1B schematically illustrate a plane view and a top view for a step of fabricating a semiconductor package in accordance with some embodiments of the disclosure.

(3) FIG. 2A and FIG. 2B schematically illustrate a plane view and a top view for a step of fabricating a semiconductor package in accordance with some embodiments of the disclosure.

(4) FIG. 3A and FIG. 3B schematically illustrate a plane view and a top view for a step of fabricating a semiconductor package in accordance with some embodiments of the disclosure.

(5) FIG. 4A and FIG. 4B schematically illustrate a plane view and a top view for a step of fabricating a semiconductor package in accordance with some embodiments of the disclosure.

(6) FIG. 5A and FIG. 5B schematically illustrate a plane view and a top view for a step of fabricating a semiconductor package in accordance with some embodiments of the disclosure.

(7) FIG. 6 schematically illustrates a cross sectional view of a semiconductor package in accordance with some embodiments of the disclosure.

(8) FIG. 7 schematically illustrate a plane view for cutting traces of a precut step of fabricating a semiconductor package in accordance with some embodiments of the disclosure.

(9) FIG. 8 schematically illustrate a cross sectional view for cutting depths of a precut step of fabricating a semiconductor package in accordance with some embodiments of the disclosure.

(10) FIG. 9 schematically illustrates a semiconductor package in accordance with some embodiments of the disclosure.

(11) FIGS. 10A to 10D schematically illustrate respective enlarged portions of a semiconductor package in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

(12) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(13) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(14) Embodiments discussed herein may be discussed in a specific context, namely a package component is having one or more integrated circuit dies. In some embodiments, the package component is a system-on-integrated-substrate (SoIS) package. The package component includes an interconnect structure bonded to a redistribution structure. In some embodiments, other components such as a locate interconnect component, an integrated voltage regulator, an integrated

passive device, a static random-access-memory, the like, or a combination thereof can optionally be embedded in the redistribution structure. The embodiments disclosed herein may be applied to various systems such as high performance computing (HPC), edge computing, cloud computing, data centers, networking, and artificial intelligence. The redistribution structure is connected to one or more semiconductor device such as the integrated circuit dies and provides electrical connection between the integrated circuit dies and an interconnect structure such as a core substrate and/or between the integrated circuit dies. The core substrate is additionally connected to a set of external conductive features. In such a manner, the integrated circuit dies are electrically connected to the core substrate, and ultimately to the external conductive features, through the core substrate and the redistribution structure. Conductive connectors used to connect the core substrate to the redistribution structures may take the form of, for example, a ball grid array (BGA). Integration of such conductive connectors may provide flexibility in placement for semiconductor devices, such as integrated passive device (IPD) chips, integrated voltage regulators (IVRs), active chips, among other electrical components, to implement system-on-a-chip (SoC) type of package components, thus reducing fabrication complexity. Such embodiments may also provide a greater amount of flexibility for various other package configurations as well. Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

(15) FIG. 1A and FIG. 1B schematically illustrate a plane view and a top view for a step of fabricating a semiconductor package in accordance with some embodiments of the disclosure. In FIG. 1A and FIG. 1B, a redistribution structure **110** is formed in advance and is transferred on a carrier **200**. The carrier **200** may be a film carrier supported by a frame **202**. In some embodiments, the redistribution structure **110** may have a circular wafer-like shape and be arranged in multiple package regions **102**. FIG. 1B may specifically present an exemplary cross section of the redistribution structure **110** of FIG. 1A taken along line I-I while omit other portions. In some embodiments, an adhesive layer (not shown) may be formed on the carrier **200** to facilitate a subsequent debonding of the carrier **200**. In addition, a plurality of conductive connectors **120** may be disposed on the redistribution structure **110** prior to transferring the redistribution structure **110** onto the carrier **200** and the conductive connectors **120** may be positioned between the redistribution structure **110** and the carrier **200**. In some embodiments, the conductive connectors **120** may be formed on the redistribution structure **110** after the redistribution structure **110** is separated from the carrier **200**.

(16) In some embodiments, the redistribution structure **110** includes insulating layers **112** and redistribution layers **114** disposed between the insulating layers **112**. The insulating layers **112** and the redistribution layers **114** may be alternately disposed in the thickness direction. In some embodiments, two redistribution layers **114** may be separated by one of the insulating layers **112**. The insulating layers **112** may be formed of one or more suitable dielectric materials such as prepreg, resin coated copper (RCC), an oxide (e.g., silicon oxide), a nitride (e.g., silicon nitride), a photo image dielectric (PID), a polymer material such a PBO, a photosensitive polymer material, a molding material, a polyimide material, a low-k dielectric material, another dielectric material, the like, or a combination thereof. The insulating layer **208A** may be formed by a process such as lamination, coating, (e.g., spin-coating), CVD, the like, or a combination thereof. The insulating layers **112** may have a thickness of between about 5 μm and about 50 μm , such as about 15 μm , although any suitable thickness may be used. In some embodiments, the insulation layers **112** may be patterned to form with openings exposing the underlying redistribution layer **114** using suitable

photolithographic mask and etching process. In some embodiments in which one or more of the insulating layers **112** is formed of a photosensitive polymer such as PBO, polyimide, BCB, or the like, the openings may be patterned directly using an exposure and development process.

(17) In some embodiments, the redistribution layers **114** are formed by a conductive material such as copper, titanium, tungsten, aluminum, another metal, the like, or a combination thereof. In some embodiments, the redistribution layers **114** may be formed by initially forming a seed layer (not shown) of one or more thin layers of a conductive material that aids in the formation of a thicker layer during subsequent processing steps. The seed layer may comprise a layer of titanium created using processes such as sputtering, evaporation, or PECVD processes, depending upon the desired materials. A photoresist (also not shown) may then be formed and patterned to cover the seed layer using, e.g., a spin coating technique. Once the photoresist has been formed and patterned, a conductive material may be formed on the seed layer through a deposition process such as electroplating or electroless plating, or the like. Once the conductive material has been formed, the photoresist may be removed through a suitable removal process such as ashing or chemical stripping. Additionally, the portions of the seed layer that were covered by the photoresist may be removed through, for example, a suitable wet etch process or dry etch process, which may use the conductive material as an etch mask. The remaining portions of the seed layer and conductive material form the redistribution layers **114**. One of the redistribution layers **114** may extend in the opening of an underlying insulating layer **114** to be connected to another of the redistribution layers **114** to accomplish required electrical paths. The redistribution structure **110** may have one or more blank zones **104** and the redistribution layers **114** are not arranged in the blank zones **104**. The blank zones **104** may be arranged at the boundary between the package regions **102**, but the disclosure is not limited thereto.

(18) In some embodiments, contact pads **116** may be formed at a side of the redistribution structure **110** and contact pads **118** may be formed at an opposite of the redistribution structures **110**. The contact pads **116** are positioned between the conductive connectors **120** and the redistribution structure **110** under the orientation of the redistribution structure **110** shown in FIG. **1B** and the contact pads **118** are revealed at a side away from the carrier **200** and thus the contact pads **118** are presented in the plane view of FIG. **1A**. The contact pads **116** are connected to corresponding one of the redistribution layers **114** in the redistribution structure **110** through openings formed in corresponding one of the insulation layers **112**. Similarly, the contact pads **118** are connected to corresponding one of the redistribution layers **114** in the redistribution structure **110** through openings formed in corresponding one of the insulation layers **112**. The conductive connector **120** is disposed on the contact pads **116** to be electrically connected correspondingly to the redistribution structure **110**. In some embodiments, the fabricating method of the contact pads **116** and the contact pads **118** may include the fabricating method of the redistribution layers **114** and the contact pads **116** and the contact pads **118** may be considered as the outermost ones of the redistribution layer **114**. In some embodiments, the contact pads **116** and the contact pads **118** may be fabricated through a different method of fabricating the redistribution layers **114**.

(19) In some embodiments, an optional solderability treatment (e.g., pre-soldering treatment) may be performed on the exposed surfaces of the contact pads **116** prior to forming the conductive connectors **120**. The conductive connectors **120** may be, for example, contact bumps or solder balls (e.g., C4 balls, ball grid array (BGA)), although any suitable type of connectors may be utilized. In some embodiments in which the conductive connectors **120** are contact bumps, the conductive connectors **120** may include a material such as tin, or other suitable materials, such as silver, lead-free tin, or copper. In some embodiments in which the conductive connectors **120** are tin solder bumps, the conductive connectors **120** may be formed by initially forming a layer of tin using such a technique such as evaporation, electroplating, printing, solder transfer, ball placement, etc. Once a layer of tin has been formed on the structure, a reflow may be performed in order to shape the material into the desired bump shape for the conductive connectors **120**.

(20) In some embodiments, the redistribution structure **110** is fabricated in advance on a substrate (not shown) and the conductive connectors **120** are optionally formed on the redistribution structure **110** while the redistribution structure **110** remains on the substrate. Subsequently, a transferring process is adopted for transferring the redistribution structure **110** on the carrier **200** from the substrate to obtain the structure shown in FIG. 1A and FIG. 1B. In some embodiments, the conductive connectors **120** may be formed on the redistribution structure **110** after the redistribution structure **110** is separated from the carrier **200**. As shown in FIG. 1A and FIG. 1B, the conductive connectors **120** as well as the redistribution structure **110** are arranged in multiple package regions **102** each of which is predetermined to be formed a semiconductor package. In FIG. 1A, five package regions **102** are shown as an example and the quantity of the package regions **102** may be determined based on the required designs.

(21) FIG. 2A and FIG. 2B schematically illustrate a plane view and a top view for a step of fabricating a semiconductor package in accordance with some embodiments of the disclosure. In FIG. 2A and FIG. 2B, a pre-cut process is performed. FIG. 2B may specifically present an exemplary cross section of the redistribution structure **110** of FIG. 2A taken along line II-II while omit other portions. In some embodiments, the pre-cut process is performed by carving the redistribution structure **110** along cutting traces CT1 using a dicing tool **300**. In some embodiments, the dicing tool **300** may be a blade, a saw, a laser drill, an etching apparatus, and the like, or combinations thereof. The redistribution structure **110** may have one or more blank zones **104** and the cutting traces CT1 may be planned within the blank zones **104**. In some embodiments, the blank zones **104** may be absent of the redistribution layers **114**, the contact pads **116** and the contact pads **118**.

(22) In some embodiments, the pre-cut process is performed by carving the redistribution structure **110** until a cutting depth CD is reached to form reduced structures **110R** in the redistribution structure **110**. In some embodiments, each of the reduced structures **110R** may extend through one or more of the insulating layers **114** of the redistribution structure **110**. The pre-cut process is controlled so that the cutting depth CD do not reach the carrier **200** and thus the carrier **200** is not damage under the pre-cut process. In some embodiments, the cutting depth CD is smaller than the total thickness of the redistribution structure **110** so that one or more of the insulation layers **112** is not cut to maintain the continuity of the redistribution structure **110**. In some embodiments, one or more wedged edge WE may be formed through the pre-cut process, and each of the reduced structures **110R** is defined and demarcated by the corresponding wedged edge WE. The wedged edge WE may be extend obliquely with respect to the thickness direction of the redistribution structure **110**. Accordingly, a portion of the insulating layers **112** of the redistribution structure **110** that is cut through the pre-cut process may have a gradually shrink edge in the thickness direction away from the carrier **200**.

(23) In some embodiments, the cutting traces CT1 may extend within the blank zones **104** positioned at the boundary between the package regions **102**, but the disclosure is not limited thereto. In some embodiments, the cutting traces CT1 may be planned within the package regions **102** and may be planned along a linear trace, a circular tract, a zigzag trace, a dash trace, or other alternative traces. In some embodiments, the redistribution layers **114** in the redistribution structure **110** may be absent in the blank zones **104** and the reduced structure **110R** is located in the blank zones **104**, so that the redistribution layers **114** in the redistribution structure **110** are spaced from the reduced structure **110R**, and particularly, spaced from the wedged surface WE defining the reduced structure **110R**. In some embodiments, the blank zones **104** may not extend through the thickness of the redistribution structure **110** so that one or more of the redistribution layers **114** between the bottom level of the blank zone **104** and the level of contact pads **116** may extend laterally across the reduced structure **110R**. For example, one or more of the insulating layers **112** may be disposed between one of the redistribution layers **114** and the bottom of the reduced structure **110R** when the cutting depth CD is smaller than the thickness of the redistribution

structure **110**, so that an electric connection path may be established laterally across the reduced structure **110R**. In some embodiments, the pre-cut process may be performed without cutting the conductive material of the redistribution layers **114** since the redistribution layers **114** are absent in the blank zones **104**. Accordingly, the conductive material of the redistribution layers **114** is not exposed at the wedged edges **WE** and the electric connection paths established by the redistribution layers **114** are not damage by the pre-cut process.

(24) FIG. 3A and FIG. 3B schematically illustrate a plane view and a top view for a step of fabricating a semiconductor package in accordance with some embodiments of the disclosure. FIG. 3B may specifically present an exemplary cross section of the redistribution structure **110** of FIG. 3A taken along line III-III while omit other portions. In FIG. 3A and FIG. 3B, conductive connectors **130** are formed on the redistribution structure **110** that is pre-cut. The conductive connectors **130** are disposed on and connected to the contact pads **118** on the surface of the redistribution structure **110** opposite to the carrier **200**. The conductive connectors **130** may be ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive connectors **130** may include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the conductive connectors **130** are formed by initially forming a layer of solder through evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once a layer of solder has been formed on the structure, a reflow may be performed in order to shape the material into the desired bump shapes. In another embodiment, the conductive connectors **130** include metal pillars (such as a copper pillar) formed by a sputtering, printing, electro plating, electroless plating, CVD, or the like. The metal pillars may be solder free and have substantially vertical sidewalls. In some embodiments, a metal cap layer is formed on the top of the metal pillars. The metal cap layer may include nickel, tin, tin-lead, gold, silver, palladium, indium, nickel-palladium-gold, nickel-gold, the like, or a combination thereof and may be formed by a plating process. In some embodiments, a diameter of the conductive connectors **130** may be in a range of 25 um to 800 um, for example.

(25) FIG. 4A and FIG. 4B schematically illustrate a plane view and a top view for a step of fabricating a semiconductor package in accordance with some embodiments of the disclosure. FIG. 4B may specifically present an exemplary cross section of a portion of the redistribution structure **110** of FIG. 4A taken along line IV-IV while omit other portions. In FIG. 4A and FIG. 4B, interconnect structures **140** are bonded to the redistribution structure **110** at a plurality of package regions **102**. Each of the interconnect structures **140** is disposed on one of the package regions **102**. In some embodiments, the interconnect structures **140** may be bonded to the redistribution structure **110** through the conductive connectors **130**.

(26) In some embodiments, each of the interconnect structures **140** may be, for example, an interposer or a “semi-finished substrate” which could either have active and passive devices or else may be free from active and passive devices. The interconnect structures **140** can also provide stability and rigidity to the attached redistribution structure **110**. The interconnect structures **140** may be bonded to the redistribution structure **110** at respective package regions **102**. During bonding the interconnect structures **140**, a bonding stress directing toward the carrier **200** may be applied to the redistribution structure **110** and may be concentrated at the respective package regions **102**, which is possible to cause undesirable warpage of the redistribution structure **110** and the carrier **200** since the carrier **200** is a film carrier with flexibility and the interconnect structures **140** are more rigid structures. In the embodiment, the redistribution structure **110** is pre-cut to have the reduced structures **110R** which break the continuity of one or more insulating layers **112** in the redistribution structure **110** so that the reduced structures **110R** may provide an effect of releasing the bonding stress of the respective interconnect structures **140**. For example, during bonding the interconnect structures **140** to the redistribution structure **110**, the bonding stress may be at least

partially released and/or reduced through the reduced structure **110R**. As such, the bonding stress may not concentrate at certain regions and the warpage due to the bonding stress may be mitigated. Accordingly, the connection between the interconnect structures **140** at the respective package regions **102** to the corresponding conductive connectors **130** may be ensured since the warpage effect is mitigated. In some embodiments, the reduced structures **110R** may be deformed, for example narrowed, during bonding the interconnect structure **140** to buffer the bonding stress, but the disclosure is not limited thereto.

(27) In some embodiments, the interconnect structure **140** may include a core layer **142** and two build-up layers **144** and **146** respectively located on two surfaces of the core layer **142**. In some embodiments, the core layer **142** includes a core dielectric layer **142A**, conductive lids **142B** and **142C**, and plated through holes **142D**. In some embodiments, the core dielectric layer **142A** includes prepreg (which contains epoxy, resin, silica filler and/or glass fiber), Ajinomoto Buildup Film (ABF), resin coated copper foil (RCC), polyimide, photo image dielectric (PID), ceramic core, glass core, molding compound, a combination thereof, or the like. However, the disclosure is not limited thereto, and other dielectric materials may also be used. The core dielectric layer **142A** may be formed by a lamination process, a coating process, or the like. The conductive lids **142B** and **142C** are formed on the opposite sides of the core dielectric layer **142A**. In some embodiments, the conductive lids **142B** and **142C** include copper or other suitable conductive material, for example.

(28) In some embodiments, the plated through holes **142D** are disposed in and penetrate through the core dielectric layer **142A**, which provide electrical connection between the conductive lids **142B** and **142C**. In some embodiments, the plated through holes **142D** may be lined with a conductive material and filled up with an insulating material. In some embodiments, the method of forming the plated through holes TH includes the following operations. First, through holes (not shown) are formed at the predetermined positions by, for example, a mechanical or laser drilling, an etching, or another suitable removal technique. A desmear treatment may be performed to remove residues remaining in the through holes. Subsequently, the through holes may be plated with one or more conductive materials to a predetermined thickness, thereby providing the plated through holes **142D**. For example, the through holes may be plated with copper with an electroplating or an electroless plating.

(29) In some embodiments, the conductive lids **142B** and **142C**, and the plated through holes **142D** may be formed by the following steps. First, first conductive material (not shown) is respectively formed on two opposite surfaces of the core dielectric layer **142A**. Then, the plated through holes **142D** are formed to penetrate the core dielectric layer **142A** as mentioned before and provide electrical connection between the first conductive material respectively formed on both surfaces of the core dielectric layer **142A**. Thereafter, second conductive material is respectively formed over the first conductive materials on the opposite surfaces of the core dielectric layer **142A**, where the second conductive material may be different from the first conductive material. In some embodiments, the first and second conductive materials may be formed by using any suitable method (e.g., chemical vapor deposition (CVD) sputtering, printing, plating, or the like). Then, the first conductive materials and the second conductive materials may be patterned together to form core conductive layers (not shown) and the conductive lids **142B** and **142C** respectively. In some embodiments, the first and second conductive materials may be partially removed using a photolithography and etching process or another suitable removal technique.

(30) Each of the build-up layers **144** and **146** may have any suitable number of dielectric layers **144A** and routing layers **144B**, including more or fewer than that shown in FIG. 4B. In some embodiments, one or both of the build-up layers **144** and **146** may be omitted. In some embodiments, the number of layers of the build-up layer **144** may be different from the number of layers of the build-up layer **146**. In some embodiments, the build-up layers **144** is formed by forming the dielectric layer **144A** over the conductive lids **142B** and the core layer **142A**. In some

embodiments, the dielectric layer **144A** may be a material such as a build-up material, ABF, a prepreg material, a laminate material, another material similar to those described above for the core layer **142**, the like, or combinations thereof. The dielectric layer **144A** may be formed by a lamination process, a coating process, or another suitable process. In some embodiments, openings are formed in the dielectric layer **144A** that expose portions of the conductive lids **142B** for subsequent electrical connection. In some embodiments, the openings are formed by, for example, an etching process, a laser drilling technique, or the like. Other processes, e.g., mechanical drilling or the like, may also be used in other embodiments. In some embodiments, an optional surface preparation process (e.g., a desmear process or the like) may be performed after the openings are formed.

(31) A conductive material is then deposited to form routing layer **144B** on the dielectric layer **144A** and within the openings in the dielectric layer **144A**. In some embodiments, the routing layer **144B** is formed by first forming a seed layer and a patterned mask over the dielectric layer **144A**. The patterned mask may be, for example, a patterned photoresist layer. Openings in the patterned mask may expose portions of the seed layer on which conductive material will subsequently be formed. The conductive material may then be deposited on the exposed regions of the dielectric layer **144A** and within the openings in the dielectric layer **144A** using, for example, a plating process, an electroless plating process, or another process. After depositing the conductive material, the patterned mask layer (e.g., the photoresist) may be removed using a wet chemical process or a dry process (e.g., an ashing process). In this manner, an additional routing layer (e.g., routing layer **144A**) is formed over and electrically connected to the conductive lid **142B**. Additional dielectric layers and routing layers may then be formed adjacent to the routing layer **144B** and dielectric layer **144A** to provide additional routing along with electrical connection within the build-up layer **144**. The dielectric layers **144A** and the routing layers **144B** may be formed in alternating layers, and the build-up layer **144** may have any suitable number and configuration of dielectric layers and routing layers. In some embodiments, dielectric layers **146A** and the routing layers **146B** may be formed adjacent to the conductive lid **142C** to form the build-up layer **146**. The build-up layer **146** may be formed using a process similar to that of the build-up layer **144**, described above. However, any suitable process may be utilized.

(32) In some embodiments, protection layers **148A** and **148B** are formed over the build-up layers **144** and **146** of the interconnect structure **140**. The protection layers **148A** and **148B** may be e.g., a solder resist material or a PBO material, and may be formed to protect the surfaces of the build-up layers **144** and **146**. In some embodiments, each of the protection layers **148A** and **148B** may be a photosensitive material formed by printing, lamination, spin-coating, or the like. The photosensitive material may then be exposed to an optical pattern and developed, forming openings in the photosensitive material. In other embodiments, the protection layers **148A** and **148B** may be formed by depositing a non-photosensitive dielectric layer (e.g., silicon oxide, silicon nitride, the like, or a combination), forming a patterned photoresist mask over the dielectric layer using suitable photolithography techniques, and then etching the dielectric layer using the patterned photoresist mask using a suitable etching process (e.g., wet etching or dry etching). The protection layers **148A** and **148B** may be formed and patterned over the build-up layers **144** and **146** using the same techniques. Other processes and materials may also be used.

(33) FIG. 5A and FIG. 5B schematically illustrate a plane view and a top view for a step of fabricating a semiconductor package in accordance with some embodiments of the disclosure. FIG. 5B may specifically present an exemplary cross section of a portion of the redistribution structure **110** of FIG. 5A taken along line V-V while omit other portions of the redistribution structure **110**. In FIG. 5A and FIG. 5B, an insulating material **150** is formed over the carrier **202** to encapsulate the package substrate **150**. In some embodiments, the insulating material **150** may be molding compound, molded underfill, polymer such as polyimide, polybenzoxazole (PBO) or benzocyclobutene (BCB), Ajinomoto Buildup Film (ABF) or other suitable encapsulating

materials, for example. In some embodiments, the insulating material **150** may be formed by a molding process or other suitable methods. In the exemplary embodiment, the insulating material **150** is formed by an over-molding process. In some embodiments, the insulating material **150** may cover a sidewall of the interconnect structure **140**. In addition, the insulating material **150** is formed between the outermost conductive patterns of the build-up layer **146**, formed aside the conductive connectors **130** and the outermost conductive patterns of the redistribution structure **110**.

Accordingly, the interconnect structure **140** is embedded in the insulating material **150**.

(34) In FIG. 5A and FIG. 5B, the insulation material **150** may extend to fill the reduced structure **110R** of the redistribution structures **110** and thus the wedged surface WE of the redistribution structure **110** is covered by and in contact with the insulation material **150**. In some embodiments, the insulation material **150** may completely fill the reduced structure **110R** and thus there might be no void formed inside reduced structure **110R**. Accordingly, the redistribution structure **110** is filled to have desirable mechanical property. The insulation material **150** may include a first portion **152** laterally surrounding the interconnect structure **140**, a second portion **154** filling the reduced structure **110R** of the redistribution structure **110** and a third portion **156** filling the space between the conductive connectors **130**. The first portion **152**, the second portion **154** and the third portion **156** of the insulation material **150** may be formed in a one-piece form with the same material. In some embodiments, the third portion **156** may be formed using a first insulation material firstly and the first portion **152** and the second portion **154** may be subsequently formed using a second insulation material. Therefore, the insulation material **150** may be formed by dual-step encapsulation process. In the dual-step encapsulation process, the first insulation material may be the same as or different from the second material based on different requirements.

(35) As shown in FIG. 5B, the sidewall **S142** of the core layer **142**, the sidewall **S144** of the build-up layer **144** and the sidewall **S146** of the build-up layer **146** may form the sidewall **S140** of the interconnect structure **140**. The first portion **152** of the insulation material **150** is in contact with the sidewalls **S142**, **S144** and **S146** to at least laterally encapsulate the interconnect structure **140**. The second portion **154** of the insulation material **150** may be in contact with the insulating layers **112** of the redistribution structure **110** extending to the wedged surface WE. The second portion **154** of the insulation material **150** may have a taper shape along with the wedged edge WE. The third portion **156** may be disposed in the gap between the interconnect structure **140** and the redistribution structure **110** and surround the conductive connectors **130**.

(36) After the insulation material **150** is formed to encapsulate the interconnect structure **140**, a singulation process is performed to separate the package regions **102**. In some embodiments, the singulation process may include separating the structures at the respective package regions **102** using a dicing tool **302** which may be similar to the dicing tool **300** described in FIG. 3B. In some embodiments, the separated structures obtained from dicing the structure of FIG. 5B are, for example, package substrates and the separated package substrate may be utilized to fabricate a semiconductor package **100** as shown in FIG. 6. In some embodiments, after the structure of FIG. 5B is singulated to individual package substrates, the singulated package substrates are oriented upside down with respect to the orientation shown in FIG. 5B and the conductive connectors **120** may be fabricated on the individual package substrates after the package substrates are singulated. In some embodiments, the conductive connectors **120** may be omitted from the fabrication of the package substrates.

(37) Specifically, FIG. 6 shows the semiconductor package **100** that is fabricated from one package substrate **160** singulated from the structure in one package region **102** of FIG. 5B. In some embodiments, after the singulation process, a semiconductor device **170** is bonded onto the package substrate **160**, an underfill **180** is formed to fill the gap between the package substrate **160** and the semiconductor device **170**, and external connectors **190** are formed on the package substrate **160** opposite to the semiconductor device **170**, so that the semiconductor package **100** is obtained. Accordingly, the semiconductor package **100** includes at least the package substrate **160**

and the semiconductor device **170** bonded to the package substrate **160**. The semiconductor device **170** may be a packaged device and may be bonded to the package substrate **160** through the conductive connector **120**. In some embodiments, the semiconductor package **100** may be mounted to an external substrate (not shown) through the external connectors **190** to obtain a system package. The external substrate may be a printed circuit board such as a laminate substrate formed as a stack of multiple thin layers (or laminates) of a polymer material such as bismaleimide triazine (BT), FR-4, ABF, or the like.

(38) The package substrate **160** is fabricated by the steps described in FIGS. **1A** to **5A** and **1B** to **5B**. The package substrate **160** may include a redistribution structure **110**, an interconnect structure **140** and an insulation material **150**. In some embodiments, the conductive connectors **120** and the conductive connectors **130** are disposed at two opposite sides of the redistribution structure **110**. The interconnect structure **140** may be connected to the redistribution structure **110** through the conductive connectors **130** between the redistribution structure **110** and the interconnect structure **140**, and the semiconductor device **170** is bonded to the redistribution structure **110** through the conductive connectors **120**. Therefore, the redistribution structure **110** may be positioned between the semiconductor device **170** and the interconnect structure **140** in the thickness direction. In some embodiments, the insulation material **150** laterally surrounds the redistribution structure **110** and the interconnect structure **140** so that the sidewall **E160** of the package substrate **160** may be defined by the sidewall **E150** of insulation material **150**. In some embodiments, the sidewall **E110** of the redistribution structure **110** may be partially exposed without being completely covered by the insulation material **150** so that the sidewall **E160** of the package substrate **160** may include the sidewall **E150** and an exposed portion of the sidewall **E110**.

(39) In some embodiments, the conductive conductors **120** are arranged with a pitch of about 130 μm , the conductive connectors **130** are arranged with a pitch of about 500 μm , and the external connectors **190** are arranged with a pitch of about 1000 μm . In some embodiments, the conductive conductors **120** have a dimension of about 70 μm , the conductive connectors **130** have a dimension of about 250 μm , and the external connectors **190** have a dimension of about 500 μm . Accordingly, the package substrate **160** enlarges the scale of the bonding sites to facilitate the bonding to an external substrate such as a circuit board or the like. However, the values described herein are examples and should not construed as limitation of the disclosure.

(40) The redistribution structure **110** includes insulating layers **112** and redistribution layers **114** alternately disposed in the thickness direction. The redistribution layers **114** includes a plurality of conductive patterns laterally surrounded by the insulating layers **112** and the redistribution layers **114** are absent at and spaced from the sidewall **E110** of the redistribution structure **110**. Side edges **E114** of the insulating layers **112** form the sidewall **E110** and the side edges **E114** may be immediately connected to one another without being interposed. In some embodiments, no material of the redistribution layers **114** is exposed at the sidewall **E110**. In some embodiments, the sidewall **E110** of the redistribution structure **110** may include the wedged surface **WE** formed by the pre-cut process described in FIG. **2A** and FIG. **2B**. In some embodiments, the pre-cut process reaches a depth that extends through the thickness of the redistribution structure **110** so that an entire of the sidewall **E110** forms the wedged surface **WE**. In some embodiments, the pre-cut process reaches a depth that is smaller than the thickness of the redistribution structure **110** so that a portion of the sidewall **E110** forms the wedged surface **WE** while the other portion of the sidewall **E110** may be coplanar with the sidewall **E150** of the insulation material **150**. The redistribution structure **110** may have a lateral dimension gradually changed in the thickness direction to form the wedged surface **WE**. For example, the lateral dimension of the redistribution structure **110** may be gradually reduced in a direction from the conductive connectors **120** to the conductive connectors **130**, or gradually reduced in a direction directing toward the interconnect structure **140**. Therefore, the wedged surface **WE** may be gradually shrunk in the direction from the conductive connectors **120** to the conductive connectors **130**, or in the direction directing toward the interconnect structure

140. In some embodiments, the insulation material **150** may compensate the oblique configuration of the wedged surface **WE** to obtain a substantially upright sidewall **E150**.

(41) The interconnect structure **140** is connected to the redistribution structure **110** through the conductive connectors **130**. The interconnect structure **140** may have a lateral dimension smaller than a maximum dimension of the redistribution structure **110**. In some embodiments, the sidewall **E140** of the interconnect structure **140** may be laterally shrunk with respect to at least a portion of the sidewall **E110** of the redistribution structure **110**. The interconnect structure **140** includes a core layer **142** and two build-up layers **144** and **146**. The core layer **142** and the two build-up layers **144** and **146** may have the same lateral dimension so that the interconnect structure **140** may have an upright structure. Sidewalls of the core layer **142** and the two build-up layers **144** and **146** constructing the sidewall **E140** of the interconnect structure **140** may be covered by and in contact with the insulation material **150**. The external connectors **190** are disposed on the interconnect structure **140** at an external bonding surface **B140** away from the redistribution structure **110**. The external bonding surface **B140** is not covered by the insulation material **150**.

(42) The insulation material **150** may include a first portion **152** laterally surrounding the interconnect structure **140**, a second portion **154** laterally surrounding the wedged surface **WE** of the redistribution structure **110** and a third portion **156** filling the gap between the redistribution structure **110** and the interconnect structure **140** and spaces between the conductive connectors **130**. Accordingly, the insulation material **150** laterally surrounds and encapsulates the interconnect structure **140**, the conductive connectors **130** and the redistribution structure **110**. The first portion **152**, the second portion **154** and the third portion **156** may be formed in a one-piece form, but the disclosures is not limited thereto. In some embodiments, one or more of the first portion **152**, the second portion **154** and the third portion **156** may be an individually formed portion. The insulation material **150** may include a material such as a molding compound, an epoxy, an underfill, a dispense molding underfill (DMUF), a resin, or the like. The insulation material **150** may be dispensed using, e.g., a molding process, such as a transfer molding process, an injection process, combinations of these, or the like. The insulation material **150** can protect the conductive connectors **130** and can provide structural support for the interconnect structure **140** as well as the redistribution structure **110**.

(43) The sidewall **E150** of the insulation material **150** is formed by the singulation process including dicing the structure shown in FIG. 5B along the boundary between adjacent package regions **102**. In some embodiments, the lateral dimensions of the redistribution structure **110** and the interconnect structure **140** are different to form a stair-like structure and the insulation material **150** is formed laterally surround the redistribution structure **110** and the interconnect structure **140** to compensate the stair-like structure to obtain a substantially upright structure of the package substrate **160**. In some embodiments, the sidewall **E160** of the package substrate **160** may extend linearly along the thickness direction.

(44) The first portion **152** of the insulation material **150** is in contact with and covers the sidewall **E140** of the interconnect structure **140**. The first portion **152** may laterally surround the interconnect structure **140** to form an encircled ring pattern in a bottom view (not shown). The first portion **152** may expose the external bonding surface **B140** of the interconnect structure **140** so that the external connectors **190** disposed on the external bonding surface **B140** of the interconnect structure **140** are exposed without being encapsulated. In some embodiments, the first portion **152** of the insulation material **150** may be leveled with the external bonding surface **B140** of the interconnect structure **140**. For example, the end surface of the first portion **152** may be coplanar with the exposed surface of the protection layer **148A** in the interconnect structure **140**.

(45) In some embodiments, the second portion **154** of the insulation material **150** may extend along the wedged surface **WE** of the redistribution structure **110**. The second portion **154** may have a width **W154** measured in the lateral direction. The width **W154** of the second portion **154** may be altered along with the wedged surface **WE** of the redistribution structure **110**. In some

embodiments, the width **154** of the second portion **154** may be gradually reduced in a direction away from the interconnect structure **140**. The second portion **154** may have a taper structure in the cross section as shown in FIG. **6** to compensate the wedged surface WE. In addition, the second portion **154** may fill the reduced structure **R110** and thus an extending depth **D154** of the second portion **154** may be corresponding to the cutting depth CD of the pre-cut process described in FIG. **2A** and FIG. **2B**.

(46) The third portion **156** of the insulation material **150** fills the gap between the redistribution structure **110** and the interconnect structure **140** and surrounds the conductive connectors **130** to encapsulate the conductive connectors **130**. The third portion **156** of the insulation material **150** may be formed in a one-piece form with the first portion **152** and the second portion **154**. In some embodiments, the third portion **156** may be formed in an individual piece independent from the first portion **152** and the second portion **154**. In some embodiments, the third portion **156** may be formed of a material independent from the first portion **152** and the second portion **154**.

(47) The semiconductor device **170** may include one or more devices, which may include devices designed for an intended purpose such as a memory die (e.g., a DRAM die, a stacked memory die, a high-bandwidth memory (HBM) die, etc.), a logic die, a central processing unit (CPU) die, an I/O die, a system-on-a-chip (SoC), a component on a wafer (CoW), an integrated fan-out structure (InFO), a package, the like, or a combination thereof. In some embodiments, the semiconductor device **170** includes integrated circuit devices, such as transistors, capacitors, inductors, resistors, metallization layers, external connectors, and the like, therein, as desired for a particular functionality. In some embodiments, the semiconductor device **170** may include more than one of the same type of device, or may include different devices. In some embodiments, the semiconductor device **170** may be a packaged semiconductor device.

(48) In some embodiments, the semiconductor device **170** is placed on the package substrate **160** through a pick up and place process, but the disclosure is not limited thereto. In some embodiments, after the semiconductor device **170** is placed on the conductive connectors **120** on the package substrate **160**, a reflow process may be adopted to bond the semiconductor device **170** to the conductive connectors **120** on the package substrate **160**. Accordingly, the semiconductor device **170** may be electrically connected to the redistribution structure **110** of the package substrate **160** through the conductive connectors **120**. In some embodiments, external connectors are formed on the semiconductor device **170** instead of or in addition to the conductive connectors **120** formed on the redistribution structure **110**. In some embodiments, the conductive connectors **120** are omitted, and the semiconductor device **170** is bonded to the redistribution structure **110** of the package substrate **160** using a direct bonding technique such as thermocompression bonding, hybrid bonding, metal-to-metal bonding, or the like. However, any suitable bonding technique may be utilized.

(49) In some embodiments, the underfill **180** may be deposited along the gap between the semiconductor device **170** and the redistribution structure **110**. The underfill **180** may also at least partially surround some conductive connectors **120**. The underfill **180** may be a material such as a molding compound, an epoxy, an underfill, a molding underfill (MUF), a resin, or the like. In some embodiments, the underfill **180** may extend to partially cover the sidewall **E170** of the semiconductor device **170**. In some embodiments, further semiconductor device may be bonded to the package substrate **110**, and thus the semiconductor package **100** may include two or more semiconductor devices bonded to the package substrate **160**. Accordingly, the structure of FIG. **6** including one semiconductor device **170** is an exemplary example and is not intended to limited the disclosure.

(50) The semiconductor package **100** includes the package substrate **160** forming by bonding the interconnect structure **140** to the redistribution structure **110** and the redistribution structure **110** has one or more reduced structure **R110** filled with the insulation material **150**. The insulation material **150** may extend into the redistribution structure **110** in a finger-like structure in the cross section to

cover the wedged surface WE defining the reduced structure **110R**. The finger-like structure of the insulation material **150** may be the second portion **154** in FIG. **6**. The second portion **154** of the insulation material **150** may have a taper shape with non-fixed width **W154**. The second portion **154** is formed by filling the reduced structures **110R** formed in the step of FIG. **2A** and FIG. **2B** and thus the second portion **154** may have a plane view shape extending along the cutting trace **CT1**. In the embodiment, the plane view shape of the second portion **154** may be an encircled ring-like pattern surrounding the package substrate **110**, but the disclosure is not limited thereto.

(51) FIG. **7** schematically illustrate a plane view for cutting traces of a pre-cut step of fabricating a semiconductor package in accordance with some embodiments of the disclosure. The structure shown in FIG. **7** may provide non-limited alternative examples of the arrangement of the cutting traces of FIG. **2A** and thus the same reference numbers in FIGS. **2A** and **7** may refer the same elements. In FIG. **7**, the redistribution structure **110** is carried by the carrier **200** supported by the frame **202**. The redistribution structure **110** may include a plurality of package regions **102**. In FIG. **7**, a pre-cut process may be performed as the step indicated in FIG. **2A**. The pre-cut process in the embodiment of FIG. **7** may be performed along the cutting traces **CT1**, cutting traces **CT2**, cutting traces **CT3** and cutting traces **CT4**. In some embodiments, the cutting traces **CT1** may be planned along the boundaries of adjacent package regions **102** as described in FIG. **2A**. In some embodiments, the cutting traces **CT2** may be planned to pass through one or more of the package regions **102** in one direction and the cutting traces **CT2** may be planned to pass through one or more of the package regions **102** in another direction. In some embodiments, the pitches of the cutting traces **CT2** or **CT3** may be variable or identical. In some embodiments, one or more of the cutting traces **CT2** or **CT3** may pass through one package region **102**. In some embodiments, the cutting traces **CT4** may be planned to form an encircled ring-like trace with the package regions **102**.

(52) In some embodiments, the redistribution structure **110** may have blank zones (referring the blank zone **104** in FIG. **2B**) and the cutting traces **CT1**, **CT2**, **CT3** and **CT4** are arranged within the blank zones. Therefore, the pre-cut process may not cut the redistribution layers (referring the redistribution layers **114** in FIG. **2B**) in the redistribution structure **110**. Though FIG. **7** shows that the cutting traces **CT1**, **CT2**, **CT3** and **CT4** are all arranged in one structure, but the disclosure is not limited thereto. In various embodiments, the cutting traces **CT1**, **CT2**, **CT3** and **CT4** may be respectively applied to different structures. For example, one or more embodiment may include the structure of the cutting traces **CT1** arranged in the redistribution structure **110** without other cutting traces, one or more embodiment may include the structure of the cutting traces **CT2** arranged in the redistribution structure **110** without other cutting traces, one or more embodiment may include the structure of the cutting traces **CT3** arranged in the redistribution structure **110** without other cutting traces, and one or more embodiment may include the structure of the cutting traces **CT4** arranged in the redistribution structure **110** without other cutting traces. In some embodiments, the redistribution structure **110** may be pre-cut along one or more selected from the cutting traces **CT1**, **CT2**, **CT3** and **CT4**.

(53) In some embodiments, the pre-cut process performed along the cutting traces **CT1**, **CT2**, **CT3** and **CT4** would form corresponding reduced structures in the redistribution structure **110**. In addition, as shown in the step of FIG. **5A** and FIG. **5G**, the insulation material **150** may be formed to fill the reduced structures. Therefore, in the fabricated package substrate **160**, the second portion **154** filling the reduced structures may have a plane view pattern as shown in any of the cutting traces **CT1**, **CT2**, **CT3** and **CT4**. For example, in some embodiments adopting the cutting trace **CT4** in the pre-cut process, the second portion **154** of the insulation material **150** may have a ring-like plane view within the redistribution structure **110** of the package substrate **160**.

(54) FIG. **8** schematically illustrate a cross sectional view for cutting depths of a pre-cut step of fabricating a semiconductor package in accordance with some embodiments of the disclosure. The structure shown in FIG. **8** may provide non-limited alternative examples of the cut depth of FIG.

2B and thus the same reference numbers in FIGS. 2B and 7 may refer the same elements. In FIG. 8, the structure of the redistribution structure **110** subjected to the pre-cut process described in FIG. 2A and FIG. 2B is shown. The redistribution structure **110** may include one or more blank zones selected from blank zones **104A** to **104C** where the redistribution layers **114** are absent and the insulation layers **112** are immediately stacked on one another without being interposed. The blank zones **104A** to **104D** may reach different levels of the redistribution structure **110**. For example, in FIG. 8, the blank zones **104C** and **104D** may not extend through the thickness of the redistribution structure **110** while the blank zones **104A** and **104B** extend through the thickness of the redistribution structure **110**. One or more of the blank zones **104A** to **104D** may be applied in one embodiment of the redistribution structure **110**. In other words, the redistribution structure **110** may include one of the blank zones **104A** to **104D** without other blank zones **104A** to **104D**, two of one of the blank zones **104A** to **104D** without the other blank zones **104A** to **104D**, or the like. In some embodiments, the blank zones **104A** to **104D** may be applied in different embodiments of the redistribution structure **110**.

(55) In some embodiments, the pre-cut process is performed along the blank zone **104A** until a cutting depth CD1 reaches to form a reduced structure **110RA**. The cutting depth CD1 may be substantially the same as the thickness of the redistribution structure **110** and thus, the reduced structure **110RA** may extend through all insulation layers **112** of the redistribution structure **110**. Accordingly, the wedged surface WE1 defining and demarcating the reduced structure **110RA** may extend through the thickness of the redistribution structure **110**. The wedged surface WE1 may be gradually shrunk in the direction away from the conductive connectors **120**.

(56) In some embodiments, the pre-cut process is performed along the blank zone **104B** until a cutting depth CD2 reaches to form a reduced structure **110RB**. The cutting depth CD2 may be substantially the same as the thickness of the redistribution structure **110** and thus, the reduced structure **110RB** may extend through all insulation layers **112** of the redistribution structure **110**. Accordingly, the wedged surface WE2 defining the reduced structure **110RB** may extend through the thickness of the redistribution structure **110**. The wedged surface WE2 may be gradually shrunk in the direction away from the conductive connectors **120**. In some embodiments, the wedged surface WE1 may be positioned at a boundary of the package region **102** and the wedged surface WE2 may be positioned within the package region **102**.

(57) In some embodiments, the pre-cut process is performed along the blank zone **104C** until a cutting depth CD3 reaches to form a reduced structure **110RC**. The cutting depth CD3 may be smaller than the thickness of the redistribution structure **110** and thus, the reduced structure **110RC** may reach one of the insulation layers **112** in the redistribution structure **110**. In some embodiments, the wedged surface WE3 defining the reduced structure **110RC** may stop at the one of the insulation layers **112** in the redistribution structure **110**. The wedged surface WE3 may be gradually shrunk in the direction away from the conductive connectors **120**.

(58) In some embodiments, the pre-cut process is performed along the blank zone **104D** until a cutting depth CD4 reaches to form a reduced structure **110RD**. The cutting depth CD4 may be smaller than the thickness of the redistribution structure **110** and thus, the reduced structure **110RD** may reach one of the insulation layers **112** in the redistribution structure **110**. In some embodiments, the wedged surface WE4 defining the reduced structure **110RD** may stop at the one of the insulation layers **112** in the redistribution structure **110**. The wedged surface WE4 may be gradually shrunk in the direction away from the conductive connectors **120**. In some embodiments, the wedged surface WE3 may be positioned at a boundary of the package region **102** and the wedged surface WE4 may be positioned within the package region **102**.

(59) FIG. 9 schematically illustrates a semiconductor package in accordance with some embodiments of the disclosure. In FIG. 9 the semiconductor package **200** may be fabricated from the structure shown in FIG. 8 using the steps described in FIGS. 2A to 5A, FIGS. 2B to 5B and FIG. 6. The semiconductor package **200** may include the components similar to the semiconductor

package **100** described in FIG. **6** and thus the components indicating by the same reference numbers in the two embodiments are applicable to and incorporated to each other. The semiconductor device **200** may include a package substrate **160** including a redistribution structure **110** connected to an interconnect structure **140** and an insulation material **150** laterally encapsulating at least the interconnect structure **140**, a semiconductor device **170** bonded to the package substrate **160** and an underfill **180** fill the gap between the semiconductor device **170** and the package substrate **160**. The semiconductor device **170**, the interconnect structure **140** and the underfill **180** may refer to those described in the previous embodiments. In FIG. **9**, the redistribution structure **110** may have various reduced structures such as the reduced structures **110RA**, **110RB**, **110RC** and **110RD**. The reduced structures **110RA**, **110RB**, **110RC** and **110RD** may be formed by the precut step of forming the structure of FIG. **8**. The reduced structures **110RA**, **110RB**, **110RC** and **110RD** may be formed by various cutting depths and cutting traces of the precut process as described in FIG. **2A** and FIG. **2B**. The insulation material **150** may include the first portion **152** laterally surrounding the interconnect structure **140**, the second portions **154** filling the reduced structures **110RA**, **110RB**, **110RC** and **110RD** and the third portion **156** filling the spaces between the redistribution structure **110** and the interconnect structure **140**. In the embodiment, the second portions **154** filling the reduced structures **110RA**, **110RB**, **110RC** and **110RD** are respectively indicated by the reference numbers **154A**, **154B**, **154C** and **154D**.

(60) FIGS. **10A** to **10D** schematically illustrate respective enlarged portions of a semiconductor package in accordance with some embodiments of the disclosure. Referring to FIG. **9** and FIG. **10A**, the reduced structure **110RA** extends through the thickness of the redistribution structure **110** and through all the insulating layers **114** of the redistribution structure **110**. The reduced structure **110RA** is positioned at edge to serve as the sidewall **E110A** of the redistribution structure **110**. The second portion **154A** fills the reduced structure **110RA** and extends along the wedged surface **WE1** defining the reduced structure **110RA**. The second portion **154A** may have a lateral dimension gradually reduced in a direction away from the interconnect structure **140** to have a taper pattern, but the disclosure is not limited thereto. The sidewall **E150** of the insulation material **150** may extend linearly along the thickness direction of the package substrate **160** to compensate the oblique configuration of the wedged surface **WE1** to form the package substrate **160** with an upright structure. For example, the sidewall **E110A** as well as the wedged surface **WE1** is oblique to the sidewall **E150** which is known as the sidewall **E160**.

(61) Referring to FIG. **9** and FIG. **10B**, the reduced structure **110RB** extends through the thickness of the redistribution structure **110**. The reduced structure **110RB** is positioned between different portions of the redistribution layers **114** in the redistribution structure **110**. The second portion **154B** fills the reduced structure **110RB** and extends along the wedged surface **WE2** defining the reduced structure **110RB**. The second portion **154B** may extend through all of the insulating layers **112** and the redistribution layers **114** are spaced from the second portion **154B** of the insulation material **150**. The second portion **154B** may have a lateral dimension gradually reduced in a direction away from the interconnect structure **140** to have a taper shape between the redistribution layers **114** in the redistribution structure **110**. The second portion **154B** may form a finger-like structure extending in the redistribution structure **110** and between different portions of the redistribution layers **114**. The second portion **154B** is connected to the third portion **156** surrounding the conductive connectors **120** to have a pushpin-like shape in a cross section, but the disclosure is not limited thereto.

(62) Referring to FIG. **9** and FIG. **10C**, the reduced structure **110RC** does not extend through the thickness of the redistribution structure **110** and stops at one of the insulation layers **112** in the redistribution structure **110**. The second portion **154C** fills the reduced structure **110RC** and extends along the wedged surface **WE3** defining the reduced structure **110RC**. The second portion **154C** may have a lateral dimension gradually reduced in a direction away from the interconnect structure **140** to have a taper pattern. The reduced structure **110RC** is positioned at edge and thus the

sidewall **E110C** of the redistribution structure **110** includes the wedged surface **WE3** and a vertical surface **VE**. The wedged surface **WE3** is oblique and shrunk with respect to the vertical surface **VE**. The wedged surface **WE3** is formed by the pre-cut process described in FIGS. **2A**, **2B** and **8** while the vertical surface **VE** is formed by the singulation process described in FIGS. **5A** and **5B**. The vertical surface **VE** may be substantially vertical and may be not. The vertical surface **VE** may be coplanar with the sidewall **E150** of the insulation material **150** to form the sidewall **E160'** of the package substrate **160**. The vertical surface **VE** is not covered by the insulation material **150**. The second portion **154C** may compensate the reduced structure **110RC** and thus the sidewall **E160'** of the package substrate **160** may have an upright configuration.

(63) Referring to FIG. **9** and FIG. **10D**, the reduced structure **110RD** does not extend through the thickness of the redistribution structure **110** and stop at one of the insulation layers **112** of the redistribution structure **110**. Specifically, the reduced structure **110RD** extends through one or more of the insulating layers **112** and one or more other of the insulating layers **112** are not broken by the reduced structure **110RD**. For example, in FIG. **9**, the redistribution structure **110** includes six insulating layers **112** and the reduced structure **110RD** extends through three of the insulating layers **112** adjacent to the interconnect structure **140** while the reduced structure **110RD** stops at the fourth insulating layer **112** calculated from the side adjacent to the interconnect structure **140**, but the disclosure is not limited thereto. Accordingly, the second portion **154D** filling the reduced structure **110RD** may extend through and be in contact with the same numbers of the insulating layers **112**, and the redistribution layers **114** are spaced from the second portion **154D**. The reduced structure **110RD** is positioned between different portions of the redistribution layers **114** in the redistribution structure **110**. The second portion **154D** extends along the wedged surface **WE4** defining the reduced structure **110RD**. The second portion **154D** may have a lateral dimension gradually reduced in a direction away from the interconnect structure **140** to have a taper pattern inserted in the redistribution structure **110**. The second portion **154D** may form a finger-like structure in the redistribution structure **110** in the cross section. In some embodiments, a portion of the redistribution layers **114** in the redistribution structure **110**, such as the redistribution layer **114'** may be positioned at a level outside the extending depth of the reduced structure **110RD** and may laterally extend through opposite sides of the reduced structure **110RD**. Therefore, the redistribution layer **114'** may overlap with the second portion **154D** of the insulation material **150** in the thickness direction. In some embodiments, one or more insulating layers **112** continuously extends between the second portion **154D** filling the reduced structure **110RD** and the redistribution layer **114'** so that the redistribution layer **114'** is not in contact with the second portion **154D**.

(64) In some embodiments, the second portion **154A** filling the reduced structure **110RA** may be individually applied in one embodiment of a semiconductor package, such as the semiconductor device **100**. Similarly, the second portion **154B** filling the reduced structure **110RB**, the second portion **154C** filling the reduced structure **110RC** and the second portion **154D** filling the reduced structure **110RD** may be independently applied in various embodiments of a semiconductor package. In some embodiments of the semiconductor package, one or more of the second portion **154A** filling the reduced structure **110RA**, the second portion **154B** filling the reduced structure **110RB**, the second portion **154C** filling the reduced structure **110RC** and the second portion **154D** filling the reduced structure **110RD** may be presented in the same semiconductor package. Therefore, the semiconductor package **200** is an exemplary example showing various alternatives of the second portion **154** without limiting to include all the alternatives in one package.

(65) In some embodiments of the disclosure, the method of fabricating a semiconductor package may include a pre-cut process on a redistribution structure in a substantially wafer form to form a reduced structure defining by wedged surface. The interconnect structures are bonded to the redistribution structure at the respective package regions and the stress due to the bonding of the interconnect structures may be releases, spread and/or mitigated through the reduced structure.

Accordingly, the warpage of the carrier carrying the redistribution structure in the wafer form may be mitigated and, which improves the connections between different interconnect structures and the redistribution structure and enhances the yield rate of the method of fabricating the semiconductor package. The semiconductor package may include the package substrate having the insulation material filling the reduced structure so that the package substrate has a substantially upright structure and desirable mechanical property. In some embodiments, the reduced structure filled with the insulation material would not cause unwanted voids which facilitates to ensure the mechanical property of the package device.

(66) In light of the above, the semiconductor package may include one or more reduced structure in the redistribution structure and the insulation material encapsulating the interconnect structure may fill the reduced structure to obtain a package substrate with a substantially upright shape. The reduced structure is formed by a pre-cut process performing on the redistribution structure and facilitates the improvement of the connection between the interconnect structure and the redistribution structure. Therefore, the quality of the semiconductor package may be ensured.

(67) In accordance with an embodiment, a semiconductor package includes a package substrate and a semiconductor device. The package substrate includes a redistribution structure, an interconnect structure bonded to the interconnect structure and an insulation material laterally surrounding the interconnect structure, wherein the redistribution structure has a reduced structure and the insulation material fills the reduced structure. The semiconductor device is bonded to the package substrate. The redistribution structure includes a wedged surface defining the reduced structure. The insulation material includes a first portion laterally surrounding the interconnect structure and a second portion filling the reduced structure. The second portion has a lateral dimension gradually reduced in a direction away from the interconnect structure. The insulation material further fills a gap between the interconnect structure and the redistribution structure. The redistribution structure includes insulating layers and redistribution layers alternately disposed in a thickness direction and the reduced structure extends through one or more of the insulating layers. The redistribution structure has a blank zone absent of the redistribution layers and the reduced structure is located in the blank zone. The insulation material covers a sidewall of the redistribution structure.

(68) In accordance with another embodiment, a semiconductor package includes a package substrate and a semiconductor device. The package substrate includes a redistribution structure, an interconnect structure bonded to the interconnect structure and an insulation material laterally surrounding the interconnect structure, wherein the redistribution structure comprises insulating layers and redistribution layers between the insulating layers, and the insulation material include a portion extend through one or more of the insulating layers, and the redistribution layers are spaced from the portion of the insulation material. The semiconductor device is bonded to the package substrate. The portion of the insulation material has a taper shape. The portion of the insulation material extend through all of the insulating layers of the redistribution structure. A sidewall of the redistribution structure includes a wedged surface and a vertical surface, and the wedged surface is oblique with respect to the vertical surface. A sidewall of the insulation material is coplanar with the vertical surface. External connectors are disposed on an external bonding surface of the interconnect structure and the insulation material exposes the external bonding surface.

(69) In accordance with yet another embodiment, a method of fabricating a semiconductor package includes transferring a redistribution structure including insulating layers and redistribution layers between the insulating layers on a carrier; performing a pre-cut process to form a reduced structure in the redistribution structure; bonding interconnect structures to the redistribution structure at a plurality of package regions; forming an insulation material filling the reduced structure; and bonding a semiconductor device to the redistribution structure. The precut process is performed until a cutting depth extending one or more of the insulating layers. The precut process is performed along a cutting trace within a blank zone of the redistribution structure and the blank zone is absent of the redistribution layers. The cutting trace extends along a boundary between the

package regions. The insulation material laterally surrounds the interconnect structure. A singulation process is further performed to separate the package regions.

(70) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor package comprising: a package substrate comprising a redistribution structure, an interconnect structure bonded to the redistribution structure and an insulation material laterally surrounding the interconnect structure, wherein the redistribution structure has a reduced structure and the insulation material fills the reduced structure and further fills a gap between the interconnect structure and the redistribution structure; and a semiconductor device bonded to the package substrate.
2. The semiconductor package of claim 1, wherein the redistribution structure has a wedged surface defining the reduced structure.
3. The semiconductor package of claim 1, wherein the insulation material includes a first portion laterally surrounding the interconnect structure and a second portion filling the reduced structure.
4. The semiconductor package of claim 3, wherein the second portion has a lateral dimension gradually reduced in a direction away from the interconnect structure.
5. The semiconductor package of claim 1, wherein the redistribution structure includes insulating layers and redistribution layers alternately disposed in a thickness direction and the reduced structure extends through one or more of the insulating layers.
6. The semiconductor package of claim 5, wherein the redistribution structure has a blank zone absent of the redistribution layers and the reduced structure is located in the blank zone.
7. The semiconductor package of claim 1, wherein the insulation material covers a sidewall of the redistribution structure.
8. A semiconductor package, comprising: a package substrate comprising a redistribution structure, an interconnect structure bonded to the redistribution structure and an insulation material laterally surrounding the interconnect structure and further filling a gap between the interconnect structure and the redistribution structure, wherein the redistribution structure comprises insulating layers and redistribution layers between the insulating layers, and the insulation material include a portion extending through one or more of the insulating layers and the redistribution layers are spaced from the portion of the insulation material; and a semiconductor device bonded to the package substrate.
9. The semiconductor package of claim 8, wherein the portion of the insulation material has a taper shape.
10. The semiconductor package of claim 8, wherein the portion of the insulation material extend through all of the insulating layers of the redistribution structure.
11. The semiconductor package of claim 8, wherein a sidewall of the redistribution structure includes a wedged surface and a vertical surface, and the wedged surface is oblique with respect to the vertical surface.
12. The package of claim 11, wherein a sidewall of the insulation material is coplanar with the vertical surface.
13. The semiconductor package of claim 8, further comprising external connectors disposed on an external bonding surface of the interconnect structure and the insulation material exposes the

external bonding surface.

14. A semiconductor package comprising: a package substrate comprising a redistribution structure, an interconnect structure, conductive connectors connecting between the redistribution structure and the interconnect structure, and an insulation material laterally surrounding the conductive connectors and the interconnect structure, wherein the insulation material comprises a taper protrusion extending toward the redistribution structure and further fills a gap between the interconnect structure and the redistribution structure; and a semiconductor device bonded to the redistribution structure.

15. The semiconductor package of claim 14, wherein the taper protrusion surrounds an outer edge of the redistribution structure.

16. The semiconductor package of claim 14, wherein the taper protrusion is surrounded by the redistribution structure.

17. The semiconductor package of claim 14, wherein the taper protrusion has a protrusion height less than a thickness of the redistribution structure.

18. The semiconductor package of claim 14, wherein the taper protrusion has a lateral dimension gradually reduced in a direction away from the interconnect structure.

19. The semiconductor package of claim 14, wherein the redistribution structure has a shape complementary to the taper protrusion.

20. The semiconductor package of claim 14, wherein the conductive connectors are located in the gap between the interconnect structure and the redistribution structure.
